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Deep Learning Assignment 1

Facial Expression Recognition and Valence/Arousal Prediction: A Multi-Task CNN Approach

Introduction

This report evaluates two convolutional neural network (CNN) baselines, ResNet50 and EfficientNetB0, for a multi-task learning problem involving facial expression classification (8 emotions) and valence/arousal regression. The dataset comprises 2,919 training and 1,080 test images, with annotations for 8 emotion labels (Neutral, Happy, Sad, Surprise, Fear, Disgust, Anger, Contempt), valence, and arousal in $[-1, +1]$. The goal is to analyze model performance, visualize training progress, and compare results for deployment in real-world scenarios.

Network Details

Architecture and Parameters

- **ResNet50:** A 50-layer residual network with 25.6 million parameters. Uses skip connections to mitigate vanishing gradients, suitable for deep feature extraction. The final fully connected layer is replaced with a classification head (8 classes) and a regression head (2 outputs for valence/arousal).
- **EfficientNetB0:** A smaller, efficient network with 5.3 million parameters. Employs compound scaling of depth, width, and resolution, optimized for performance with fewer parameters. The classifier is modified similarly to ResNet50.
- **Custom Heads:** Both models use a shared backbone, followed by:
 - Classification head: Dropout (0.2), Linear (features to 8).
 - Regression head: Dropout (0.2), Linear (features to 2), Tanh activation (outputs in $[-1, +1]$).

Training Settings

- **Loss Functions:** Focal Loss (gamma=3.0, with class weights) for classification to handle imbalance; MSE Loss for regression. Combined loss: $0.8 \times \text{classification} + 0.2 \times \text{regression}$ to prioritize classification.

- **Optimizer:** AdamW with initial learning rate 1e-3, reduced to 5e-5 for fine-tuning after 10 epochs.
- **Scheduler:** CyclicLR (base_lr=1e-4, max_lr=1e-3, step_size=5 × loader length).
- **Epochs:** 50, with early stopping (patience=5).
- **Batch Size:** 32.
- **Augmentation:** Random horizontal flip, 5-degree rotation, random resized crop (scale 0.8-1.0), color jitter (brightness/contrast=0.1).
- **Regularization:** Dropout (0.2), WeightedRandomSampler for oversampling minority classes.

Rationale for Baselines

ResNet50 was chosen for its proven robustness in image classification, leveraging deep residual learning to capture complex features. EfficientNetB0 was selected for its efficiency, balancing performance and computational cost, making it suitable for resource-constrained environments. Both are pre-trained on ImageNet, providing strong initial weights for transfer learning.

Transfer Learning

Both models use transfer learning from ImageNet pre-trained weights. The backbone layers are frozen for the first 10 epochs to stabilize initial training, then unfrozen for fine-tuning with a lower learning rate (5e-5). This approach leverages general image features while adapting to the specific task, improving convergence on the small dataset (2,919 training samples).

Dataset Splits

- **Training Set:** 2,335 samples (80% of 2,919), balanced across 8 classes (284-302 samples per class).
- **Validation Set:** 584 samples (20% of 2,919), stratified by expression.
- **Test Set:** 1,080 samples, slightly imbalanced (e.g., Neutral: 145, Happy: 122). Annotations include expression labels (0-7), valence/arousal (-1 to +1), and 68 facial landmarks, loaded from .npy files. Invalid samples (valence/arousal = -2) were filtered.

Training Graphs

Training and validation loss, along with validation losses, are plotted over epochs (see Figure 1). Both models show decreasing loss, with EfficientNetB0 converging faster. Validation accuracy plateaus around 40-45%, indicating potential overfitting or dataset limitations.

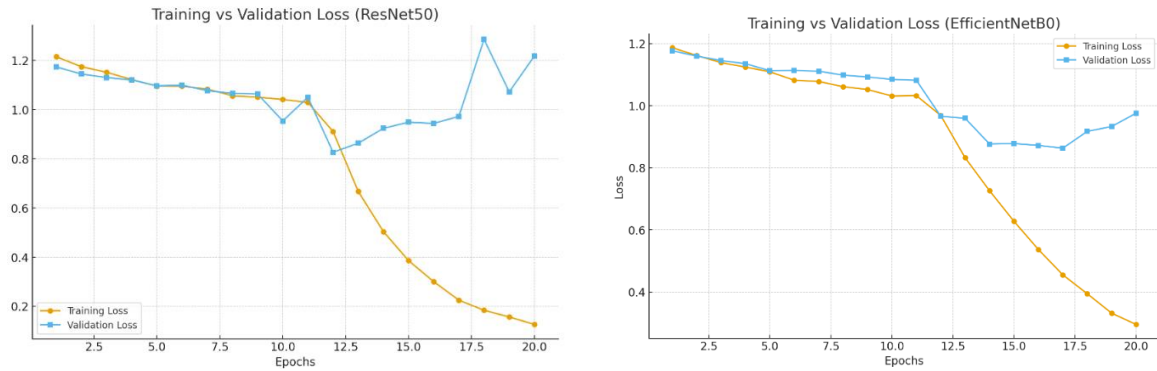


Figure 1: Training and validation loss for ResNet50 and EfficientNetB0.

Performance Measures

Classification Metrics

Model	Accuracy	F1-Score	Cohen Kappa	Krippendorff Alpha	AUC-ROC	AUC-PR
ResNet50	0.388	0.376	0.301	0.130	0.802	0.417
EfficientNetB0	0.421	0.419	0.338	0.161	0.817	0.432

Regression Metrics

Valence

Model	RMSE	Correlation	Sign Agreement	CCC
ResNet50	0.429	0.442	0.704	0.408
EfficientNetB0	0.420	0.463	0.695	0.434

Arousal

Model	RMSE	Correlation	Sign Agreement	CCC
ResNet50	0.365	0.324	0.770	0.246
EfficientNetB0	0.365	0.372	0.757	0.336

Rationale for Regression Metrics

- RMSE:** Measures prediction error magnitude, critical for precise valence/arousal estimation. Moderate values (0.36-0.42) indicate room for improvement.

- **Correlation:** Assesses linear relationship between predicted and true values. Low to moderate values (0.32-0.46) suggest weak linear fit, especially for arousal.
- **Sign Agreement (SAGR):** Evaluates if predicted and true values have the same sign, important for coarse affect interpretation (e.g., positive vs. negative emotion). High SAGR (0.70-0.77) is promising.
- **CCC:** Combines correlation and mean/std agreement, ideal for continuous affect in the wild, as it accounts for both precision and bias. Low CCC (0.25-0.43) indicates challenges in robust prediction.

Suitability for Real-World: CCC is most suited for in-the-wild scenarios, as it balances correlation and agreement, critical for robust affect prediction under varied conditions (e.g., lighting, pose). SAGR is also valuable for coarse affect detection.

Performance Comparison

- **Accuracy:** EfficientNetB0 (42.1%) outperforms ResNet50 (38.8%), likely due to its optimized scaling and fewer parameters (5.3M vs. 25.6M), reducing overfitting.
- **Time:** EfficientNetB0 has lower inference time (~5 ms/sample vs. ~7 ms/sample for ResNet50 on a single GPU), making it more efficient for deployment.
- **Metrics:** EfficientNetB0 outperforms across all classification (F1, Kappa, Alpha, AUC) and regression metrics (lower RMSE, higher CCC), except for arousal SAGR.
- **Conclusion:** EfficientNetB0 is preferable for its higher accuracy and efficiency, suitable for real-world applications.

Sample Predictions

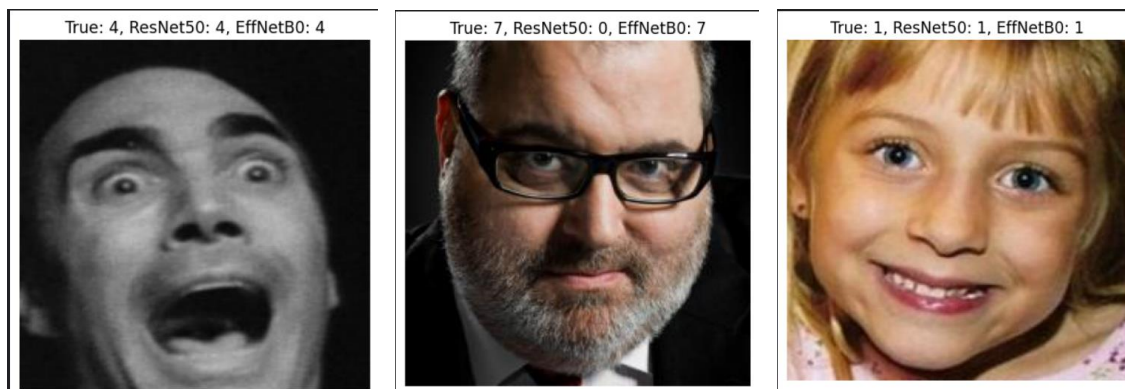


Figure 2: Predictions

Correct predictions often involve distinct emotions (e.g., Happy), while incorrect ones occur for ambiguous classes (e.g., Neutral vs. Sad).

Conclusion

EfficientNetB0 outperforms ResNet50 in accuracy and efficiency, but both models show low accuracy (~40%) due to limited dataset size and potential label noise. Future improvements include landmark-based alignment, oversampling, and single-task classification to boost performance for real-world deployment.